area like an opening in the porous materials. It occurs due to intermolecular forces between the liquid and surrounding solid surfaces. It is caused by the pressure of cohesion and adhesion, which works against gravity.

8. Viscosity

Viscosity is a physical property of fluids which shows resistance to flow. In a simple example, water has a low viscosity, as it is thin. Syrup and tar, on the other hand, have a high viscosity, as they are thick. It is denoted by the Greek letter μ (mu). It is expressed as Pa.s(Pascal second) i.e., N.s/m^2 or kg/ms.

7.2 Measurement of Pressure

7.2.1 Fluid pressure

Pressure is nothing but a force or weight. The fluid pressure at any point in a liquid is due to the weight of liquid standing above that point. Whenever a liquid (such as water, oil, etc) is in a vessel, it exerts force at all points on the side and bottom of the vessel. This fluid pressure is transmitted equally in all directions. It acts normal to the surface of contact.

7.2.2 Pascal's law

Pascal's law states that, "The pressure at any point in a static fluid (fluid at rest) acts equally in all directions".

Pressure at any point in vertical direction is only due to the weight of the liquid above that point. The liquid transmits pressure equally in all directions. Therefore, the pressure in vertical direction causes an equal horizontal pressure on the sides of the vessel.

7.2.3 Intensity of pressure

Intensity of pressure is the fluid thrust (push in a specified direction) per unit area. It is also termed as unit pressure or simply pressure. It is denoted by the letter p. If P is the force acting on an area A. Then,

$$\text{Intensity of pressure, } p = \frac{P}{A}$$

It is expressed in N/m^{2}or Pascal (Pa), kN/m^2 and N/mm^2

7.2.4 Pressure and Pressure head

Pressure at a point in a liquid is represented by the equation $p = \gamma H$

Where, γ= specific weight

H = pressure head

The vertical height of a liquid producing a particular pressure is known as the pressure head.

7.2.5 Types of pressure

The pressure may be classified as

1. Static pressure
2. Atmospheric pressure
3. Gauge pressure
4. Absolute pressure

1. Static pressure

Static pressure is the pressure that a fluid exerts when it is not moving (at rest).

2. Atmospheric pressure

Atmospheric pressure is defined as the force per unit area exerted against a surface by the weight of the air above that surface.

3. Gauge pressure

Gauge pressure is the pressure relative to atmospheric pressure. Therefore, it is positive for pressures above atmospheric pressure and negative for pressures below it. The negative gauge pressure is also called vacuum pressure.

U7.indd 138 12/11/2019 5:33:59 PM